

ViolaVision 2.0

AI-Powered Traffic Violation Intelligence Platform
Flipkart Gridlock Hackathon 2.0 | Theme 3: Computer Vision for Traffic Violations

80.5%	3-of-5	7+	SHA-256	ByteTrack
Helmet Model mAP50	Frame Confirmation	Violation Types	Evidence Integrity	Persistent Tracking

ViolaVision 2.0 is an end-to-end traffic violation detection and evidence management platform designed for deployment across Bengaluru's existing ITMS camera infrastructure. The system combines fine-tuned computer vision models with cryptographically signed evidence records, human-in-the-loop review workflows, and a real-time operations dashboard — processing continuous video feeds using the identical pipeline demonstrated on recorded traffic footage in this submission.

1. Problem Statement

Bengaluru records over 12,000 road accident fatalities annually (MoRTH 2023). The top contributing violations — helmet non-compliance, red-light running, and wrong-side driving — are largely undetected due to manual enforcement limitations. Bengaluru's existing ITMS infrastructure covers 172 junctions with 1,050+ cameras, yet automated violation detection is absent from most deployments. Manual enforcement reaches fewer than 2% of daily violation events at high-density junctions such as Silk Board, KR Circle, and Hebbal.

Key Limitations of Current Enforcement

- Manual surveillance: human officers cannot monitor hundreds of cameras simultaneously.
- No persistent evidence chain: verbal or photograph-based records are legally contestable.
- No repeat-offender intelligence: each violation is processed in isolation, missing serial offenders.
- No integration with VAHAN: owner lookup requires separate manual queries per violation.
- Single-image analysis: existing pilot systems flag violations per frame with no temporal confirmation, producing high false-positive rates.

2. Solution Architecture

ViolaVision 2.0 is structured as a seven-layer cascade, each layer building on the outputs of the previous. The architecture is frame-independent at the detection layer — the identical logic runs on a single uploaded image for testing, or on 30 frames per second from a live RTSP stream in production.

Layer 1 — Ingestion & Preprocessing

CLAHE contrast enhancement (adaptive histogram equalisation on L-channel in LAB colour space) applied to every frame before detection. Handles low-light, haze, and glare conditions common to Indian junctions at dawn/dusk without requiring IR hardware during daytime operation.

Layer 2 — Dual-Model Detection

YOLOv8s (COCO-pretrained, 80 classes) handles vehicle, motorcycle, and person detection. A separately loaded fine-tuned YOLOv8n (trained on HELMET INDIA dataset, 530 images, mAP50 = 80.5%) classifies the head region of each detected person riding a motorcycle into With Helmet or Without Helmet. The two models run independently — the base model is never replaced by the classifier, resolving the architecture flaw common to single-model approaches.

Layer 3 — Geometric Feature Engineering

Bounding box geometric features are explicitly extracted per detection: aspect ratio (w/h), normalised centroid position (norm_x, norm_y), normalised area (fraction of frame), bottom fraction (vertical position), and riding posture flag (aspect ratio within riding-posture bounds). These named features are used directly in violation rules — for example, triple-riding detection requires riding posture AND significant IoU overlap with the motorcycle bbox, preventing background pedestrians from being miscounted as riders.

Layer 4 — Temporal Confirmation via ByteTrack

YOLOv8's built-in ByteTrack assigns persistent integer IDs to each detected object across frames. A per-track violation history (deque, maxlen=5) records whether each tracked object was flagged in each processed frame. A violation is confirmed and evidence is generated only when 3 of the last 5 frames flag the same track ID for the same violation type. This eliminates single-frame false positives and makes the temporal smoothing claim genuinely real rather than architecturally described only.

Layer 5 — Cross-Modal Context Validation

Scene type (Junction / Parking Area / Highway) is used to suppress contextually impossible violations: stop-line and red-light violations are suppressed in Parking Area scenes; illegal parking detection is suppressed on Highways. In production, scene type is assigned automatically per camera ID from the ITMS camera registry. In the prototype, the operator sets it via a sidebar control, demonstrating the suppression logic.

Layer 6 — Priority Scoring

Each confirmed violation receives a weighted priority score: $P = V_w \times L_w \times T_w \times R_w$, where V_w is the violation-type weight (derived from MoRTH 2023 fatality contribution statistics — red-light and wrong-side carry weight 10.0, helmet 7.0), L_w is the location risk multiplier (Silk Board = 1.5, KR Circle = 1.4 based on accident density), T_w is the time-of-day multiplier (peak hours 7-10am and 5-8pm: 1.4, night 1.3), and R_w is the repeat-offender multiplier (first offence: 1.0, 3-6 offences: 2.0, 7+ offences: 3.0 with automatic RTO referral). Scores above 25 trigger CRITICAL auto-challan; 15-25 trigger HIGH same-day review.

Layer 7 — Evidence Generation and Active Learning

Each confirmed violation generates a tamper-evident JSON record: the record is SHA-256 hashed, then the hash is signed using RSA-PSS (2048-bit, SHA-256 digest, PSS padding). The private key is generated once and stored on disk for the prototype; production uses an Infineon SLB9670 TPM chip where the key never leaves hardware. Operator dismiss actions (with mandatory reason: poor image quality / wrong classification / no actual violation / other) are logged to a separate table, forming the retraining dataset for the active learning pipeline.

3. Violation Detection Capabilities

Violation Type	Detection Method	Severity	V_w
Helmet Non-Compliance	Fine-tuned YOLOv8n on head crop	HIGH	7.0
Triple Riding	Person count on motorcycle + posture filter	HIGH	8.0
Red-Light Violation	Stop-line cross + signal state toggle	CRITICAL	10.0
Stop-Line Violation	Vehicle bottom edge past calibrated Y	MEDIUM	6.0
Wrong-Side Driving	Position vs expected flow direction	CRITICAL	10.0
Illegal Parking	Stationary detection via track ID history	LOW	3.0
Seatbelt Non-Compliance	Architecture ready; classifier planned (post-hackathon)	HIGH	6.0

Illegal parking detection in video mode uses genuine stationary detection: each vehicle's centroid position is recorded across 15 tracked frames; a vehicle is flagged only if total centroid displacement is below 18 pixels — confirming it has not moved. This replaces the zone-heuristic approach (bottom_frac threshold) used in single-image mode, which produced false positives on aerial highway footage where all vehicles naturally appear near the bottom of the frame.

4. Model Training and Performance

Helmet Detection Classifier

Dataset: HELMET INDIA v1 (Roboflow Universe, CC BY 4.0). 530 training images, 66 validation images, 2 classes: With Helmet / Without Helmet. Images sourced from Indian roads across multiple cities, varying lighting conditions, and camera angles — providing realistic diversity for the target deployment environment.

Parameter	Value	Rationale
Base model	YOLOv8n pretrained on COCO	Transfer learning from 80-class general detector
Epochs	33 (early stopping at patience=10)	Stopped when validation mAP plateaued
Image size	640 x 640	Standard YOLOv8 input resolution
Batch size	16 (GPU)	RTX 3050 4GB VRAM — fills without OOM

Optimizer	AdamW, lr=0.001	Better generalisation than SGD on small datasets
Augmentation	Mosaic, HSV shift, flip, rotate +5deg	Simulates camera angle variation and lighting
Training time	~6 minutes (NVIDIA RTX 3050)	GPU acceleration via CUDA 12.4
mAP50 (overall)	80.5%	Industry threshold for deployment: 75%+
mAP50 (With Helmet)	91.6%	High precision on compliant riders
mAP50 (Without Helmet)	69.5%	Lower due to fewer non-compliant samples

Dual-Model Architecture Design Decision

A common mistake in helmet detection systems is replacing the base COCO detector with the fine-tuned classifier. This causes the system to lose all COCO class knowledge (motorcycle, person, car) and detect only helmet/no-helmet — making it impossible to associate a detected helmet status with a specific vehicle or rider. ViolaVision 2.0 explicitly separates the two models: YOLOv8s handles all vehicle and person detection; the helmet classifier runs only on cropped head regions of detected riders. The base model's COCO knowledge is fully preserved.

5. Evidence Integrity and Chain of Custody

Every confirmed violation generates a tamper-evident JSON evidence record. The integrity mechanism is real cryptography, not a demonstration placeholder.

Hashing and Signing Process

- All fields except the hash and signature themselves are serialised as JSON with sorted keys (deterministic ordering).
- SHA-256 hash of the serialised string is computed using Python hashlib.
- RSA-PSS signature (2048-bit key, SHA-256 digest, PSS maximum salt length) is computed over the hash using the cryptography library.
- Hash and base64-encoded signature are appended to the record.
- Tampering with any single field — including timestamp, plate number, confidence score, or GPS coordinates — produces a different hash, causing signature verification to fail with a clear error.
- A tamper simulation is built into the Evidence Verify tab: editing any field and re-verifying shows the exact failure message, demonstrating court-admissible tamper detection.

Prototype Implementation	Production Deployment
SHA-256 hashing	SHA-256 hashing (identical)
RSA-PSS software key (file on disk)	RSA-PSS + Infineon SLB9670 TPM (key never leaves hardware)
System clock timestamp	NTP-synchronised + GPS satellite dual timestamp
Mock blockchain TX ID	Polygon network real anchor transaction

Local SQLite storage	TLS 1.3 encrypted write-once forensic database
Mock VAHAN lookup	Live VAHAN BTP API integration

6. Knowledge Graph and Repeat Offender Intelligence

ViolaVision 2.0 maintains a violation graph where nodes represent vehicle plates, camera locations, and violation types, and edges represent recorded violation events. This graph enables queries that flat violation logs cannot support.

Graph Capabilities

- Repeat offender detection: vehicles with 2+ violations in 30 days are automatically flagged, with escalation to senior officer at 3+ and automatic RTO referral at 7+.
- Camera hotspot ranking: junctions are ranked by violation frequency and average priority score, guiding BBMP traffic engineering interventions.
- Plate cloning detection: a plate seen at two locations more than 5 km apart within 10 minutes is physically impossible and flags a cloned plate alert.
- Infrastructure failure signatures: violation type clustering by time and location reveals missing infrastructure (e.g. absent U-turns causing wrong-side patterns) rather than just individual offences.
- OCR confidence gating: only plates read with confidence above 0.4 and matching the Indian plate regex (two letters + two digits + one-to-three letters + three-to-four digits) are used as graph nodes, preventing low-quality reads from creating false repeat-offender connections.

The prototype implements graph queries in SQLite with a Plotly network visualisation layer. Production deployment uses Neo4j Enterprise for genuine graph traversal at city scale, with partitioned graph structure by geographic zone to support horizontal scaling across hundreds of cameras.

7. Human-in-the-Loop and Active Learning Pipeline

No automated system should have unchecked authority over enforcement actions. ViolaVision 2.0 uses a genuine human-in-the-loop architecture, not a logging afterthought — the system is structurally incapable of issuing a confirmed enforcement record without a human review step for medium-confidence violations.

Review Queue Workflow

- Every violation lands in the Review Queue with status PENDING REVIEW at the moment it is confirmed by the 3-of-5 frame threshold.
- Violations with confidence above 0.88 are auto-queued for supervisory sampling but not auto-confirmed.
- An operator must explicitly click Confirm, Dismiss, or Escalate for each violation.
- Dismissals require a mandatory reason: poor image quality / wrong classification / no actual violation / vehicle type error / other.

- Dismissal reasons are stored in a dedicated table keyed on camera ID, enabling per-camera false positive rate calculation (shown in the Analytics tab).
- When a camera's false positive rate exceeds 15%, the system automatically flags it for calibration review.

Active Learning Data Collection

The dismissal log is the active learning dataset. Every operator dismissal is a labelled negative example — the model predicted a violation, a human said it was wrong. When 500 dismissals accumulate, the system surfaces a retraining recommendation. Before any retrained model replaces the production model, it is validated in a sandboxed comparison against the current model on a held-out test set; only models that match or improve performance metrics are promoted. This prevents the active learning loop from degrading the model through noisy human labels.

8. Scalability, Deployment Roadmap, and Future Development

Deployment Architecture for Bengaluru Scale

The prototype runs as a Streamlit application on a single workstation. Production deployment adds a compute module to existing ITMS camera poles rather than replacing infrastructure — a NVIDIA Jetson AGX Orin (64GB) per junction cluster running the detection pipeline at the edge, with confirmed violations transmitted over TLS 1.3 to a central evidence server.

Phase	Timeline	Deliverable
Phase 1 (Current)	Hackathon prototype	Helmet + parking + stop-line detection on video; signed evidence; review queue; kno
Phase 2 (3 months)	Pilot: 3 junctions	Seatbelt classifier training; VAHAN BTP API integration; Neo4j deployment; ITMS R
Phase 3 (6 months)	City rollout: 50 junctions	RT-DETR upgrade; LPRNet plate reader; real blockchain anchoring; night-mode IR p
Phase 4 (12 months)	Full Bengaluru ITMS	172 junctions; active learning live retraining; accident detection (Phase 2 model); CA

Honest Assessment of Prototype Limitations

- Wrong-side driving detection in the prototype requires an operator toggle because single-image analysis cannot prove direction of motion; production uses optical flow (RAFT) across consecutive frames.
- Illegal parking requires a vehicle to remain stationary for at least 8 tracked frames in the prototype; production confirms 5+ minutes of stationarity for legal defensibility.
- Seatbelt detection architecture is defined (identical pipeline to helmet classifier, different dataset); the trained classifier is a Phase 2 deliverable due to dataset acquisition time.
- VAHAN lookup uses a mock database of 3 entries; production requires BTP API credentials from MoRTH, which require formal partnership with traffic enforcement authorities.
- Temporal confirmation uses 3-of-5 frames (approximately 0.5 seconds at 30 FPS); production tuning may adjust this threshold based on violation type and false positive rate data from the review queue.

System Capability Summary

Capability	Status	Notes
YOLOv8s object detection	BUILT	Vehicles, persons, motorcycles — COCO 80 classes
Helmet fine-tuned classifier	BUILT	mAP50 = 80.5% on HELMET INDIA dataset
Dual-model architecture	BUILT	Base + classifier run independently
ByteTrack persistent tracking	BUILT	Persistent IDs across video frames
Temporal 3-of-5 confirmation	BUILT	Real execution, not described only
Geometric feature engineering	BUILT	6 named features per bbox
Cross-modal scene validation	BUILT	Junction / Parking / Highway suppression
Stationary parking detection	BUILT	Position history per track ID
Priority scoring (4-factor)	BUILT	$V_w \times L_w \times T_w \times R_w$ formula
Serial offender detection	BUILT	Levels 1-3 + automatic RTO referral
SHA-256 + RSA-PSS signing	BUILT	Real cryptography, tamper-detectable
Tamper demonstration	BUILT	Field edit + verify fail shown live
Human-in-the-loop review	BUILT	Pending/Confirm/Dismiss/Escalate workflow
Active learning data collection	BUILT	Dismissal reasons logged per camera
Knowledge graph visualisation	BUILT	Vehicles x cameras x violations network
Plate cloning detection	BUILT	Location-time impossibility detection
CLAHE preprocessing	BUILT	LAB colour space adaptive enhancement
VAHAN owner lookup	MOCK	Real BTP API requires MoRTH credentials
Neo4j graph database	PLANNED	SQLite for prototype; Neo4j Enterprise production
Real blockchain anchoring	PLANNED	Polygon network; mock TX ID in prototype
Seatbelt classifier	PLANNED	Architecture defined; Phase 2 training
IR night-mode preprocessing	PLANNED	Hardware component; CLAHE used for demo
RTSP live stream ingestion	PLANNED	Video file used for demo reliability